



US011993581B2

(12) **United States Patent**
Poslusney et al.

(10) **Patent No.:** **US 11,993,581 B2**
(45) **Date of Patent:** **May 28, 2024**

- (54) **PYRIDAZINE COMPOUNDS FOR INHIBITING NAV1.8**
- (71) Applicant: **Lieber Institute, Inc.**, Baltimore, MD (US)
- (72) Inventors: **Michael Poslusney**, Owings Mills, MD (US); **Glen Ernst**, Bear, DE (US); **James Barrow**, Arnold, MD (US)
- (73) Assignee: **LIEBER INSTITUTE, INC.**, Baltimore, MD (US)

8,883,840 B2	11/2014	Chafeev et al.
9,783,501 B2	10/2017	Hadida-Ruah et al.
9,828,397 B2	11/2017	Anderson et al.
9,969,693 B2	5/2018	Bogdan et al.
10,000,475 B2	6/2018	Tadesse et al.
10,005,724 B2	6/2018	Andrez et al.
10,005,768 B2	6/2018	Yao et al.
2004/0110725 A1	6/2004	Dolle et al.
2009/0099233 A1	4/2009	Joshi et al.
2014/0228371 A1	8/2014	Hadida-Ruah
2019/0016671 A1	1/2019	Ahmad et al.

FOREIGN PATENT DOCUMENTS

- (*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

- (21) Appl. No.: **17/256,784**
- (22) PCT Filed: **Jul. 9, 2019**
- (86) PCT No.: **PCT/US2019/041026**
§ 371 (c)(1),
(2) Date: **Dec. 29, 2020**

CZ	140194 A3 *	12/1994		C07D 333/66
EP	0628550	12/1994		
JP	2004-238361	8/2004		
WO	WO 2009/049180	4/2009		
WO	WO 2009/049181	4/2009		
WO	WO 2009/049183	4/2009		
WO	WO 2010/049302	5/2010		
WO	WO 2010/051238	5/2010		
WO	WO 2010051238 *	5/2010		C07D 237/24
WO	WO 2011/026240	3/2011		
WO	WO 2011/140425	11/2011		
WO	WO 2012/106499	8/2012		
WO	WO 2012/112743	8/2012		
WO	WO 2012/116440	9/2012		
WO	WO 2012/125613	9/2012		
WO	WO 2013/109521	7/2013		
WO	WO 2014/120815	8/2014		
WO	WO 2014/151393	9/2014		
WO	WO 2015/010065	1/2015		
WO	WO 2018/148626	8/2018		
WO	WO 2019/014352	1/2019		

- (87) PCT Pub. No.: **WO2020/014243**
PCT Pub. Date: **Jan. 16, 2020**

- (65) **Prior Publication Data**
US 2021/0139456 A1 May 13, 2021

Related U.S. Application Data

- (60) Provisional application No. 62/695,510, filed on Jul. 9, 2018.
- (51) **Int. Cl.**
C07D 491/107 (2006.01)
C07D 237/16 (2006.01)
C07D 401/12 (2006.01)
- (52) **U.S. Cl.**
CPC **C07D 401/12** (2013.01); **C07D 237/16** (2013.01)
- (58) **Field of Classification Search**
CPC C07D 491/107; C07D 417/14;
C07D 413/14; C07D 401/12; C07D
237/04; A61K 31/553; A61K 31/541;
A61K 31/5377; A61K 31/501
USPC 540/544; 544/60, 124, 238.98;
514/211.15, 227.8, 236.5, 252.03
See application file for complete search history.

- (56) **References Cited**

U.S. PATENT DOCUMENTS

7,705,031 B2	4/2010	Wilson et al.
7,989,481 B2	8/2011	Neubert et al.
8,236,829 B2	8/2012	Neubert et al.
8,236,833 B2	8/2012	Martinborough et al.
8,309,543 B2	11/2012	Gonzalez, III et al.
8,314,125 B2	11/2012	Termin et al.
8,466,188 B2	6/2013	Chafeev et al.
8,492,403 B2	7/2013	Kawatkar et al.
8,536,195 B2	9/2013	Termin et al.

OTHER PUBLICATIONS

International Search Report and Written Opinion for PCT/US2019/041026. Mailed Oct. 28, 2019. 10 pages.

Extended European Search Report for PCT/US2019041026. Mailed Mar. 18, 2022. 11 pages.

Search Report and Written Opinion from Intellectual Property Office of Singapore for application 11202100137T. May 16, 2022. 10 pages.

International Search Report and Written Opinion for PCT/US2019/041029. Mailed Oct. 28, 2019. 16 pages.

Extended European Search Report for PCT/US2019041029. Mailed Jul. 3, 2022. 11 pages.

International Preliminary Report on Patentability for PCT/US2019/041029. Apr. 23, 2020. 6 pages.

Akopian et al., A tetrodotoxin-resistant voltage-gated sodium channel expressed by sensory neurons. *Nature*. Jan. 18, 1996;379(6562):257-62.

(Continued)

Primary Examiner — Jeffrey H Murray
(74) *Attorney, Agent, or Firm* — Casimir Jones, SC;
Jeffrey W. Childers

- (57) **ABSTRACT**

Pyridazine compounds for inhibiting Nav1.8 and methods for treating a condition, disease, or disorder associated with an increased Nav1.8 activity or expression are disclosed.

14 Claims, 2 Drawing Sheets

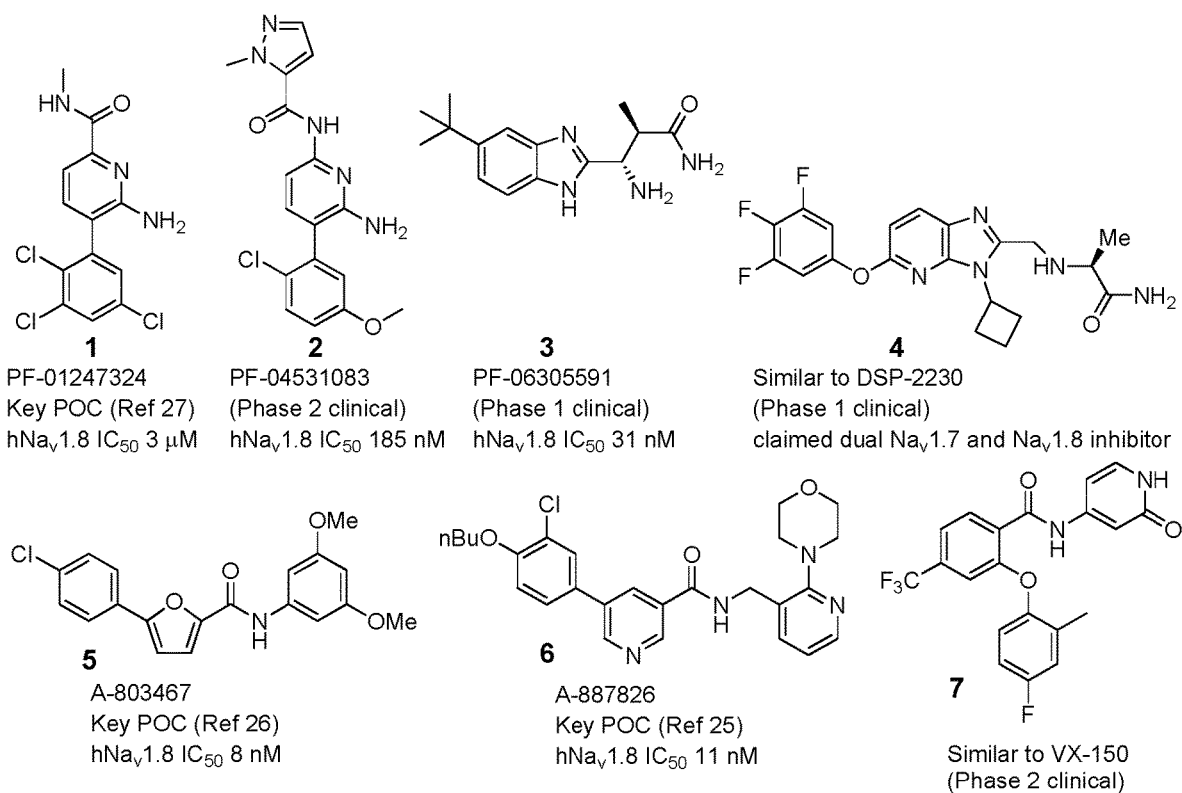


Fig. 1 (prior art)

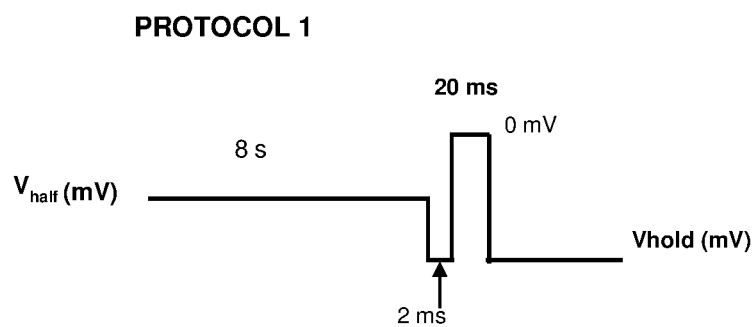
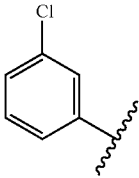
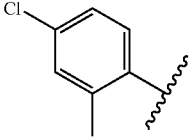
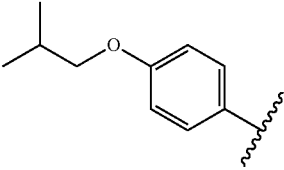
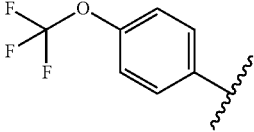
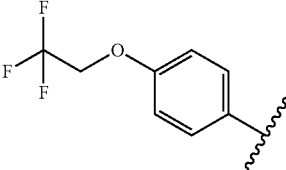
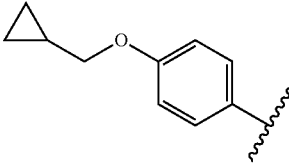
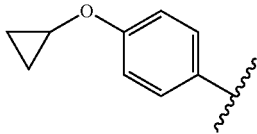
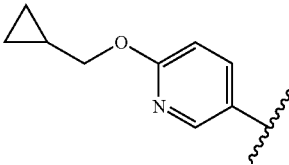
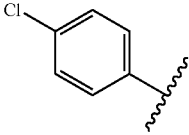


Fig. 2 (prior art)

-continued

Example Name	R	Analytical data	Preparation details
3 6-(3-chlorophenyl)-N-[(2-morpholino-3-pyridyl)methyl]pyridazine-4-carboxamide		MS, ES ⁺ m/z 410 (M + H) ⁺ .	Method A, using (3-chlorophenyl)boronic acid
4 6-(4-chloro-2-methyl-phenyl)-N-[(2-morpholino-3-pyridyl)methyl]pyridazine-4-carboxamide		MS, ES ⁺ m/z 424 (M + H) ⁺ . ¹ H-NMR (400 MHz, DMSO-d ₆) δ ppm 9.59 (d, J = 2.3 Hz, 1H), 9.51-9.57 (m, 1H), 8.22 (dd, J = 4.8, 1.7 Hz, 1H), 8.2 (d, J = 2 Hz, 1H), 7.7 (dd, J = 7.5, 1.6 Hz, 1H), 7.52-7.58 (m, 2H), 7.45-7.50 (m, 1H), 7.06 (dd, J = 7.3, 4.8, 1H), 4.58 (d, J = 5.8 Hz, 2H), 3.70-3.81 (m, 4H), 3.0-3.12 (m, 4H), 2.36 (s, 3H)	Method A, using (4-chloro-2-methylphenyl)boronic acid
5 6-(4-isobutoxyphenyl)-N-[(2-morpholino-3-pyridyl)methyl]pyridazine-4-carboxamide		MS, ES ⁺ m/z 448 (M + H) ⁺ . ¹ H-NMR (400 MHz, DMSO-d ₆) δ ppm 9.52-9.59 (m, 1H), 9.47 (d, J = 2.0 Hz, 1H), 8.51 (d, J = 2.0 Hz, 1H), 8.23 (dd, J = 4.8, 1.8 Hz, 1H), 8.15-8.21 (m, 2H), 7.71 (dd, J = 7.5, 1.9 Hz, 1H), 7.12-7.20 (m, 2H), 7.07 (dd, J = 7.6, 4.8 Hz, 1H), 4.60 (d, J = 5.8 Hz, 2H), 3.86 (d, J = 6.6 Hz, 2H), 3.71-3.81 (m, 4H), 3.03-3.12 (m, 4H), 2.00-2.14 (m, 1H), 1.02 (d, J = 6.6 Hz, 6H)	Method A, using (4-isobutoxyphenyl)boronic acid
6 N-[(2-morpholino-3-pyridyl)methyl]-6-[4-(trifluoromethoxy)phenyl]pyridazine-4-carboxamide		MS, ES ⁺ m/z 460 (M + H) ⁺ . ¹ H-NMR (400 MHz, DMSO-d ₆) δ ppm 9.53-9.62 (m, 2H), 8.62 (d, J = 2.0 Hz, 1H), 8.32-8.40 (m, 2H), 8.24 (dd, J = 4.8, 1.8 Hz, 1H), 7.72 (dd, J = 7.6, 1.8 Hz, 1H), 7.62 (dd, J = 8.8, 1.0 Hz, 2H), 7.07 (dd, J = 7.5, 4.9 Hz, 1H), 4.61 (d, J = 5.8 Hz, 2H), 3.73-3.81 (m, 4H), 3.03-3.11 (m, 4H)	Method A, using (4-(trifluoromethoxy)phenyl)boronic acid
7 N-[(2-morpholino-3-pyridyl)methyl]-6-[4-(2,2,2-trifluoroethoxy)phenyl]pyridazine-4-carboxamide		MS, ES ⁺ m/z 474 (M + H) ⁺ . ¹ H-NMR (400 MHz, DMSO-d ₆) δ ppm 9.52-9.59 (m, 1H), 9.50 (d, J = 2.0 Hz, 1H), 8.55 (d, J = 2.0 Hz, 1H), 8.20-8.27 (m, 3H), 7.72 (dd, J = 7.6, 1.8 Hz, 1H), 7.26-7.33 (m, 2H), 7.07 (dd, J = 7.6, 4.8 Hz, 1H), 4.86-4.97 (m, 2H), 4.60 (d, J = 5.8 Hz, 2H), 3.72-3.82 (m, 4H), 3.01-3.13 (m, 4H)	Method A, using (4-(2,2,2-trifluoroethoxy)phenyl)boronic acid
8 6-[4-(cyclopropylmethoxy)phenyl]-N-[(2-morpholino-3-pyridyl)methyl]pyridazine-4-carboxamide		MS, ES ⁺ m/z 446 (M + H) ⁺ . ¹ H-NMR (400 MHz, DMSO-d ₆) δ ppm 9.50-9.59 (m, 1H), 9.47 (d, J = 2.0 Hz, 1H), 8.51 (d, J = 2.0 Hz, 1H), 8.23 (dd, J = 4.8, 2.0 Hz, 1H), 8.18 (m, 2H), 7.71 (dd, J = 7.6, 1.7 Hz, 1H), 7.11-7.18 (m, 2H), 7.07 (dd, J = 7.6, 4.8 Hz, 1H), 4.6 (d, J = 5.6 Hz, 2H), 3.93 (d, J = 6.8 Hz, 2H), 3.71-3.82 (m, 4H), 3.02-3.12 (m, 4H), 1.21-1.33 (m, 1H), 0.56-0.64 (m, 2H), 0.33-0.40 (m, 2H)	Method A, using (4-(cyclopropylmethoxy)phenyl)boronic acid

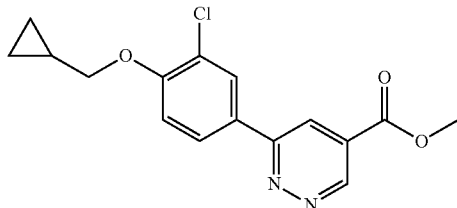
-continued

Example Name	R	Analytical data	Preparation details
15 6-[4-(cyclopropoxy)phenyl]-N-[(2-morpholino-3-pyridyl)methyl]pyridazine-4-carboxamide		MS, ES ⁺ m/z 432 (M + H) ⁺ . ¹ H-NMR (400 MHz, DMSO-d ₆) δ ppm 9.55 (t, J = 5.6 Hz, 1H), 9.48 (d, J = 2.0 Hz, 1H), 8.51 (d, J = 2.0 Hz, 1H), 8.15-8.27 (m, 3H), 7.72 (dd, J = 7.5, 1.9 Hz, 1H), 7.21-7.30 (m, 2H), 7.07 (dd, J = 7.6, 4.8 Hz, 1H), 4.60 (d, J = 5.6 Hz, 2H), 3.97 (sept., J = 3.1 Hz, 1H), 3.72-3.80 (m, 4H), 3.02-3.12 (m, 4H), 0.80-0.90 (m, 2H), 0.66-0.75 (m, 2H)	Method A, using (4-(cyclopropoxy)phenyl)boronic acid
16 6-[6-(cyclopropylmethoxy)-3-pyridyl]-N-[(2-morpholino-3-pyridyl)methyl]pyridazine-4-carboxamide		MS, ES ⁺ m/z 447 (M + H) ⁺ . ¹ H-NMR (400 MHz, DMSO-d ₆) δ ppm 9.49-9.56 (m, 2H), 8.98 (dd, J = 2.5, 0.8 Hz, 1H), 8.57 (d, J = 2.0 Hz, 1H), 8.51 (dd, J = 8.8, 2.5 Hz, 1H), 8.23 (dd, J = 4.8, 1.8 Hz, 1H), 7.72 (dd, J = 7.5, 1.9 Hz, 1H), 7.02-7.10 (m, 2H), 4.60 (d, J = 5.6 Hz, 2H), 4.21 (d, J = 7.3 Hz, 2H), 3.72-3.81 (m, 4H), 3.02-3.11 (m, 4H), 1.24-1.35 (m, 1H), 0.55-0.62 (m, 2H), 0.34-0.40 (m, 2H)	Method A, using (6-(cyclopropylmethoxy)-3-pyridyl)boronic acid
17 6-(4-chlorophenyl)-N-[(2-morpholino-3-pyridyl)methyl]pyridazine-4-carboxamide		MS, ES ⁺ m/z 410 (M + H) ⁺ .	Method A, using 4-chlorophenylboronic acid

Example 18: 6-[3-chloro-4-(cyclopropylmethoxy)phenyl]-N-[[2-(1-methylimino-1-oxo-1,4-thiazinan-4-yl)-3-pyridyl]methyl]pyridazine-4-carboxamide

Method B

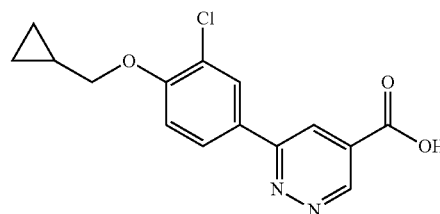
Step 1: Synthesis of methyl 6-(3-chloro-4-(cyclopropylmethoxy)phenyl)pyridazine-4-carboxylate



A microwave vial was charged with methyl 6-chloro-pyridazine-4-carboxylate (1 g, 5.8 mmol), [3-chloro-4-(cyclopropylmethoxy)phenyl]boronic acid (1.44 g, 6.37 mmol), Palladium XPhos G1 precatalyst (214 mg, 0.29 mmol), and tribasic potassium phosphate (2.46 g, 11.59 mmol). The vial was sealed, and evacuated and backfilled with nitrogen three times. THF (10 mL) and water (1 mL) were introduced via syringe, and the mixture was heated to 80° C. for about t

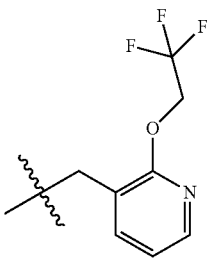
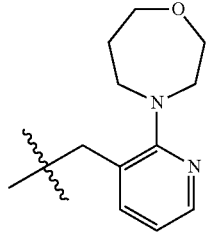
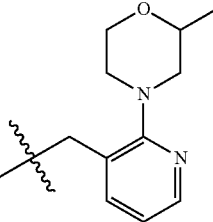
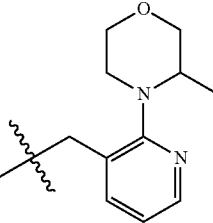
hours. The crude solution was filtered through celite, concentrated, and purified by silica gel chromatography (0-5% MeOH in DCM) to afford the desired product as a beige solid (Yield: 800 mg, 43%).

Step 2: Synthesis of 6-(3-chloro-4-(cyclopropylmethoxy)phenyl)pyridazine-4-carboxylic acid



To a vial equipped with a stir bar was added methyl 6-[3-chloro-4-(cyclopropylmethoxy)phenyl]pyridazine-4-carboxylate (1.25 g, 3.92 mmol). Methanol (15 mL) and water (2 mL) were added, followed by excess solid sodium hydroxide. After stirring at room temperature for 3 hours, the solution was diluted with water (about 100 mL) and acidified by addition of 6N HCl. The resultant precipitate was collected by filtration, rinsed with water and dried under reduced pressure to give the desired product as a solid, which was used without further purification. (Yield: 1 g, 84%).

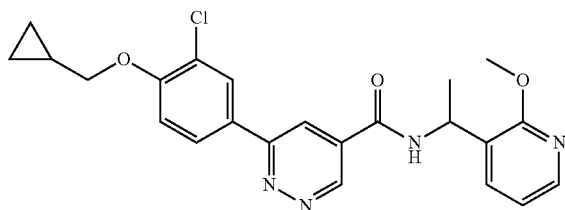
-continued

Example Name	R	Analytical data	Preparation details
20 6-[3-chloro-4-(cyclopropylmethoxy)phenyl]-N-[[2-(2,2,2-trifluoroethoxy)-3-pyridyl]methyl]pyridazine-4-carboxamide		MS, ES ⁺ m/z 493 (M + H) ⁺ . ¹ H-NMR (400 MHz, DMSO-d ₆) δ ppm 9.46-9.55 (m, 2H), 8.56 (d, J = 2.0 Hz, 1H), 8.30 (d, J = 2.3 Hz, 1H), 8.17-8.22 (m, 1H), 8.13-8.16 (m, 1H), 7.79-7.84 (m, 1H), 7.36 (d, J = 8.8 Hz, 1H), 7.13 (dd, J = 7.3, 4.8 Hz, 1H), 5.07 (quart., J = 9.1 Hz, 2H), 4.53 (d, J = 5.8 Hz, 2H), 4.05 (d, J = 6.8 Hz, 2H), 1.25-1.35 (m, 1H), 0.58-0.66 (m, 2H), 0.36-0.43 (m, 2H)	Method B, using commercially available 2-(2,2,2-trifluoroethoxy)pyridin-3-yl)methanamine in step 5
21 6-[3-chloro-4-(cyclopropylmethoxy)phenyl]-N-[[2-(1,4-oxazepan-4-yl)-3-pyridyl]methyl]pyridazine-4-carboxamide		MS, ES ⁺ m/z 496 (M + H) ⁺ . ¹ H-NMR (400 MHz, DMSO-d ₆) δ ppm 9.45-9.55 (m, 2H), 8.57 (d, J = 2.0 Hz, 1H), 8.30 (d, J = 2.3 Hz, 1H), 8.19 (dd, J = 8.8, 2.3 Hz, 1H), 8.14 (dd, J = 4.8, 2.0 Hz, 1H), 7.65 (dd, J = 7.6, 1.8 Hz, 1H), 7.35 (d, J = 8.8 Hz, 1H), 6.95 (dd, J = 7.6, 4.8 Hz, 1H), 4.56 (d, J = 5.8 Hz, 2H), 4.04 (d, J = 7.1 Hz, 2H), 3.72-3.85 (m, 4H), 3.40-3.52 (m, 4H), 1.91-2.02 (m, 2H), 1.24-1.37 (m, 1H), 0.58-0.67 (m, 2H), 0.36-0.44 (m, 2H)	Method B, using 4-oxazepane
22 6-[3-chloro-4-(cyclopropylmethoxy)phenyl]-N-[[2-(2-methylmorpholin-4-yl)-3-pyridyl]methyl]pyridazine-4-carboxamide		MS, ES ⁺ m/z 494 (M + H) ⁺ . ¹ H-NMR (400 MHz, DMSO-d ₆) δ ppm 9.47-9.57 (m, 2H), 8.57 (d, J = 1.8 Hz, 1H), 8.30 (d, J = 2.3 Hz, 1H), 8.16-8.25 (m, 2H), 7.71 (dd, J = 7.3, 1.8 Hz, 1H), 7.35 (d, J = 8.8 Hz, 1H), 7.1 (dd, J = 7.6, 4.8 Hz, 1H), 4.6 (d, J = 6.1 Hz, 2H), 4.04 (d, J = 7.1 Hz, 2H), 3.85-3.92 (m, 1H), 3.66-3.78 (m, 2H), 3.34 (s, 1H), 3.15-3.29 (m, 2H), 2.83-2.93 (m, 1H), 2.55-2.64 (m, 1H), 1.25-1.36 (m, 1H), 1.14 (d, J = 6.1 Hz, 3H), 0.58-0.66 (m, 2H), 0.36-0.43 (m, 2H)	Method B, 2-methylmorpholine
23 6-[3-chloro-4-(cyclopropylmethoxy)phenyl]-N-[[2-(3-methylmorpholin-4-yl)-3-pyridyl]methyl]pyridazine-4-carboxamide		MS, ES ⁺ m/z 494 (M + H) ⁺ . ¹ H-NMR (400 MHz, DMSO-d ₆) δ ppm 9.49-9.57 (m, 2H), 8.58 (d, J = 2.0 Hz, 1H), 8.29-8.36 (m, 2H), 8.19 (dd, J = 8.6, 2.3 Hz, 1H), 7.76 (dd, J = 7.6, 1.8 Hz, 1H), 7.35 (d, J = 8.6 Hz, 1H), 7.15 (dd, J = 7.6, 4.8 Hz, 1H), 4.64 (d, J = 6.1 Hz, 2H), 4.04 (d, J = 7.1 Hz, 2H), 3.70-3.85 (m, 3H), 3.44-3.54 (m, 1H), 3.31-3.39 (m, 1H), 3.04-3.15 (m, 1H), 2.75-2.85 (m, 1H), 1.25-1.36 (m, 1H), 0.83 (d, J = 6.32, 3H), 0.59-0.65 (m, 2H), 0.37-0.43 (m, 2H).	Method B, using 3-methylmorpholine

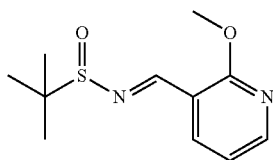
59

Example 43, 6-(3-chloro-4-(cyclopropylmethoxy)phenyl)-N-(1-(2-methoxypyridin-3-yl)ethyl)pyridazine-4-carboxamide

Method C

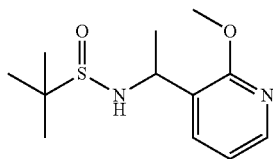


Step 1: Synthesis of (E)-N-((2-methoxypyridin-3-yl)methylene)-2-methylpropane-2-sulfinamide



2-methoxypyridine-3-carbaldehyde (2.35 g, 17.1 mmol) was added to a round-bottom flask and dissolved in DCM (25 mL). 2-methyl-2-propanesulfinamide (3.1 g, 25.7 mmol) was added, followed by titanium tetraisopropoxide (10.1 mL, 34.3 mmol). The resulting mixture was stirred at room temperature for ~18 hours, and poured into brine (about 250 mL) with vigorous stirring. The biphasic mixture was filtered through celite, and the organic layer was collected, dried and concentrated to afford the product as a yellow oil (yield: 3.1 g, 75%).

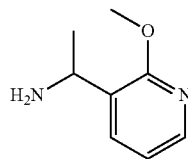
Step 2: Synthesis of N-(1-(2-methoxypyridin-3-yl)ethyl)-2-methylpropane-2-sulfinamide



(E)-N-((2-methoxypyridin-3-yl)methylene)-2-methylpropane-2-sulfinamide (531 mg, 2.2 mmol) was added to a round-bottom flask and dissolved in THF (10 mL). The mixture was cooled to -78° C., and methylmagnesium chloride (3M in THF, 1.1 mL, 3.3 mmol) was added dropwise. After stirring at -78° C. for 10 minutes, the mixture was allowed to warm to room temperature, and the reaction was quenched by the careful addition of water (about 2 mL). The mixture was diluted with dichloromethane, filtered through celite, and the filtrate was concentrated to give a colorless oil containing the desired product. This was used without further purification.

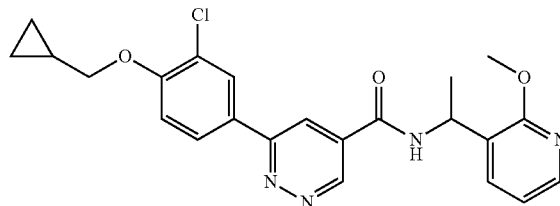
60

Step 3: Synthesis of 1-(2-methoxy-3-pyridyl)ethanamine



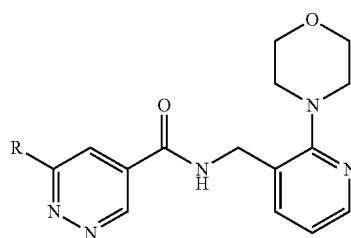
A solution of N-(1-(2-methoxypyridin-3-yl)ethyl)-2-methylpropane-2-sulfinamide (566 mg, 2.2 mmol) in DCM (10 mL) was treated with 4M HCl in dioxane (1.66 mL, 6.63 mmol) and allowed to stir at room temperature for 1.5 hours. Methanol (5 mL) was added to dissolve solids, and volatiles were evaporated under reduced pressure to give a yellow solid. This was re-dissolved in saturated sodium bicarbonate solution (25 mL) and extracted with dichloromethane. Organic layers were collected, dried, and concentrated under reduced pressure, affording a light yellow oil, which was used without further characterization.

Step 4: Synthesis of 6-(3-chloro-4-(cyclopropylmethoxy)phenyl)-N-(1-(2-methoxypyridin-3-yl)ethyl)pyridazine-4-carboxamide (Example 43)



To a suspension of Intermediate B (50 mg, 0.16 mmol) in DCM (2.5 mL) was added trimethylamine (0.046 mL, 0.73 mmol) and HBTU (68.5 mg, 0.18 mmol) and the resulting mixture was stirred at room temperature for 2 hours. 1-(2-methoxy-3-pyridyl)ethanamine (30 mg, 0.2 mmol) was added as a solution in DCM (1 mL), and stirring was continued for an additional hour. The mixture was filtered through a plug of basic alumina, concentrated, and purified by preparative HPLC (10-100% ACN/water gradient, 0.1% TFA modifier) to afford the desired product as a solid (43 mg, 60%). ES⁺ m/z 439.0 [M+H]⁺. ¹H NMR (400 MHz, DMSO-d₆) δ ppm 9.45 (d, J=2.02 Hz, 1H), 8.57 (d, J=2.02 Hz, 1H), 8.33 (d, J=2.27 Hz, 1H), 8.21 (dd, J=8.84, 2.27 Hz, 1H), 8.09 (dd, J=5.05, 1.77 Hz, 1H), 7.36 (d, J=8.59 Hz, 1H), 7.00 (dd, J=7.33, 4.80 Hz, 1H), 4.05 (d, J=6.82 Hz, 2H), 3.94 (s, 3H), 1.48 (d, J=7.07 Hz, 3H), 0.63 (dd, J=8.08, 1.77 Hz, 2H), 0.41 (dd, J=4.67, 1.64 Hz, 2H) Example 44; 6-[3-chloro-4-(cyclobutylmethoxy)phenyl]-N-[(2-morpholino-3-pyridyl)methyl]pyridazine-4-carboxamide

The following compounds have been synthesized according to Method D:



5

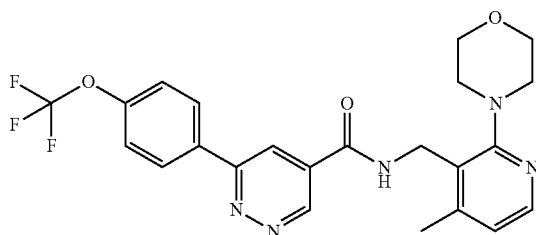
10

Example Name	R	Analytical data	Preparation details
45 6-[3-chloro-4-(1-cyclopropylethoxy)phenyl]-N-[(2-morpholino-3-pyridyl)methyl]pyridazine-4-carboxamide		MS, ES+ m/z 494 (M + H) ⁺ . ¹ H-NMR (400 MHz, DMSO-d ₆) δ ppm 9.47-9.58 (m, 2H), 8.56 (d, J = 1.8 Hz, 1H), 8.28 (d, J = 2.3 Hz, 1H), 8.23 (dd, J = 4.8, 1.8 Hz, 1H), 8.16 (dd, J = 8.6, 2.3 Hz, 1H), 7.72 (dd, J = 7.6, 1.8 Hz, 1H), 7.39 (d, J = 9.1 Hz, 1H), 7.07 (dd, J = 7.6, 4.8 Hz, 1H), 4.6 (d, J = 5.8 Hz, 2H), 4.17-4.27 (m, 1H), 3.71-3.83 (m, 4H), 3.0-3.12 (m, 4H), 1.37 (d, J = 6.1 Hz, 3H), 1.12-1.22 (m, 1H), 0.54 (dt, J = 7.9, 1.4 Hz, 2H), 0.30-0.44 (m, 2H).	Method D, using (1-bromoethyl)cyclopropane
46 6-[4-(cyclopropylmethoxy)-3-fluoro-phenyl]-N-[(2-morpholino-3-pyridyl)methyl]pyridazine-4-carboxamide		MS, ES+ m/z 464 (M + H) ⁺ . ¹ H-NMR (400 MHz, DMSO-d ₆) δ ppm 9.46-9.57 (m, 2H), 8.55 (d, J = 2.0 Hz, 1H), 8.23 (dd, J = 4.8, 1.8 Hz, 1H), 8.1 (dd, J = 12.8, 2.1 Hz, 1H), 8.04 (dt, J = 8.6, 1.1 Hz, 1H), 7.72 (dd, J = 7.6, 2.0 Hz, 1H), 7.37 (t, J = 8.7 Hz, 1H), 7.07 (dd, J = 7.6, 4.8 Hz, 1H), 4.60 (d, J = 5.6 Hz, 2H), 4.02 (d, J = 7.1 Hz, 2H), 3.72-3.81 (m, 4H), 3.01-3.12 (m, 4H), 1.23-1.34 (m, 1H), 0.57-0.65 (m, 2H), 0.34-0.41 (m, 2H).	Method D, using 2-fluoro-4-nitrophenol
47 6-[4-(cyclopropylmethoxy)-3-(trifluoromethyl)phenyl]-N-[(2-morpholino-3-pyridyl)methyl]pyridazine-4-carboxamide		MS, ES+ m/z 514 (M + H) ⁺ . ¹ H-NMR (400 MHz, DMSO-d ₆) δ ppm 9.50-9.61 (m, 2H), 8.61 (d, J = 2.0 Hz, 1H), 8.42-8.52 (m, 2H), 8.23 (dd, J = 4.8, 1.8 Hz, 1H), 7.72 (dd, J = 7.6, 1.8 Hz, 1H), 7.49 (d, J = 9.6 Hz, 1H), 7.08 (dd, J = 7.5, 4.9 Hz, 1H), 4.61 (d, J = 5.8 Hz, 2H), 4.13 (d, J = 6.8 Hz, 2H), 3.72-3.81 (m, 4H), 3.0-3.12 (m, 4H), 1.21-1.33 (m, 1H), 0.53-0.64 (m, 2H), 0.33-0.43 (m, 2H).	Method D, using 4-nitro-2-(trifluoromethyl)phenol

75

To a round-bottom flask were added 6-chloropyridazine-4-carboxylic acid (325 mg, 2.05 mmol) and dichloromethane (5 mL). Oxalyl chloride (270 μ L, 3.08 mmol) and DMF (1 drop) were added, and the mixture was stirred at room temperature for 1.5 hours. Triethylamine (572 μ L, 4.1 mmol) and (4-methyl-2-morpholino-3-pyridyl)methanamine (425 mg, 2.05 mmol) were added dropwise as a solution in dichloromethane (5 mL). The crude mixture was concentrated and purified by silica gel chromatography (eluting with a 0-5% methanol in DCM gradient) to afford the desired product (539 mg, 1.55 mmol, 76% yield) as a colorless solid. MS, ES⁺m/z 348 (M+H)⁺.

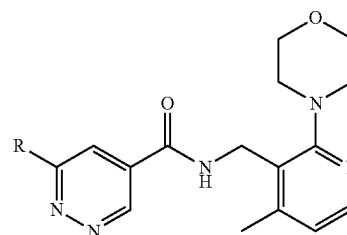
Step 4: N-[(4-methyl-2-morpholino-3-pyridyl)methyl]-6-[4-(trifluoromethoxy)phenyl]pyridazine-4-carboxamide



76

To a microwave vial equipped with a magnetic stir bar were added 6-chloro-N-[(4-methyl-2-morpholino-3-pyridyl)methyl]pyridazine-4-carboxamide (100 mg, 0.29 mmol), 4-trifluoromethoxyphenylboronic acid (89 mg, 0.43 mmol), [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium (II) (23 mg, 0.03 mmol), and cesium carbonate (187 mg, 0.58 mmol). THF (2.5 mL) and water (0.25 mL) were added, the vial was sealed, and the mixture was heated to 160° C. under microwave irradiation for 10 minutes. The crude mixture was filtered through celite, concentrated and purified by preparative reversed-phase HPLC (high pH method) to afford the desired product (41 mg, 0.09 mmol, 30% yield) as a solid. MS, ES⁺m/z 474 (M+H)⁺. ¹H NMR (500 MHz, DMSO-d₆) δ ppm 2.39 (s, 3H) 2.98-3.12 (m, 4H) 3.69-3.81 (m, 4H) 4.57 (d, J=5.34 Hz, 2H) 6.92 (d, J=7.70 Hz, 1H) 7.55-7.66 (m, 3H) 8.35 (d, J=8.80 Hz, 2H) 8.60 (d, J=1.89 Hz, 1H) 9.50 (t, J=5.42 Hz, 1H) 9.57 (d, J=1.73 Hz, 1H).

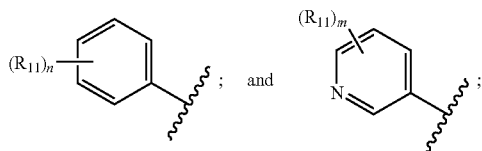
The following examples have been synthesized according to Method H:



Example Name	R	Analytical data	Prep details
60 6-(4-chlorophenyl)-N-[(4-methyl-2-morpholino-3-pyridyl)methyl]pyridazine-4-carboxamide		MS, ES ⁺ m/z 400 (M + H) ⁺ ¹ H NMR (500 MHz, DMSO-d ₆) δ ppm 2.39 (s, 3 H) 2.99-3.11 (m, 4 H) 3.69-3.80 (m, 4 H) 4.56 (d, J = 5.50 Hz, 2 H) 6.92 (d, J = 7.70 Hz, 1 H) 7.59 (d, J = 7.55 Hz, 1 H) 7.69 (m, J = 8.49 Hz, 2 H) 8.25 (m, J = 8.65 Hz, 2 H) 8.54-8.63 (m, 1 H) 9.50 (t, J = 5.27 Hz, 1 H) 9.55 (d, J = 1.73 Hz, 1 H)	Method H, using 4-chlorophenylboronic acid

87

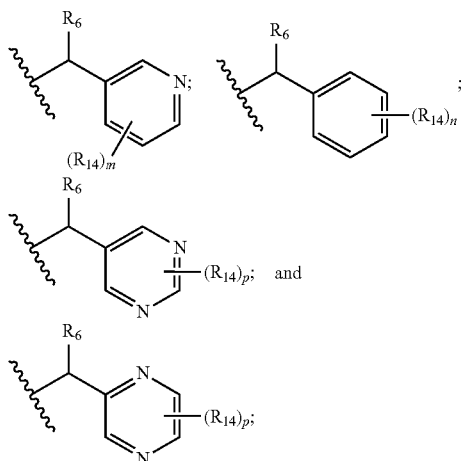
wherein:

 R_1 is selected from the group consisting of:

wherein:

 n is an integer selected from 0, 1, 2, 3, 4, and 5; m is an integer selected from 0, 1, 2, 3, and 4;

each R_{11} is independently selected from the group consisting of H, C_1 - C_8 alkyl, halogen, cyano, C_1 - C_8 alkoxy, C_1 - C_8 haloalkyl, $-O-(CH_2)_p-R_y$, wherein p is an integer selected from 0, 1, 2, 3, and 4, and R_y is selected from C_3 - C_{10} cycloalkyl and C_1 - C_8 haloalkyl, and $-O-CH(R_{12})-R_{13}$, wherein R_{12} is H or C_1 - C_8 alkyl and R_{13} is C_3 - C_8 cycloalkyl;

 R_2 is selected from the group consisting of:

wherein:

 n is an integer selected from 0, 1, 2, 3, 4, and 5; m is an integer selected from 0, 1, 2, 3, and 4; p is an integer selected from 0, 1, 2, and 3;each R_6 is independently hydrogen or C_1 - C_8 alkyl;each R_{14} is independently selected from the groupconsisting of halogen, C_1 - C_8 alkyl, C_1 - C_8 alkoxy, C_1 - C_8 haloalkyl, cyano, $-O-(CH_2)_p-R_y$, wherein p is an integer selected from 0, 1, 2, 3, and 4, and R_y is selected from C_3 - C_{10} cycloalkyl and C_1 - C_8 haloal-kyl, $-S(O)R'$, $-S(O_2)R'$, $-NR_7R_8$, where R_7 and R_8 are independently selected from C_1 - C_8 alkyl and C_1 - C_8 haloalkyl, $-NR_9R_{10}$, where R_9 and R_{10}

together form a saturated heterocyclic ring selected

from the group consisting of piperidinyl, pyrrolidi-

nyl, azetidyl, morpholinyl, thiomorpholinyl, 1,4-

oxazepanyl, 2-oxa-5-azabicyclo[2.2.1]heptanyl, and

2-oxa-6-azaspiro[3.3]heptan-6-yl, each of which can

optionally substituted with deuterium, C_1 - C_8 alkyl,

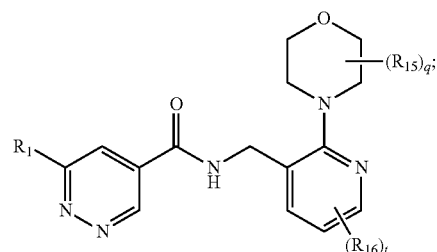
or halogen;

and pharmaceutically acceptable salts thereof.

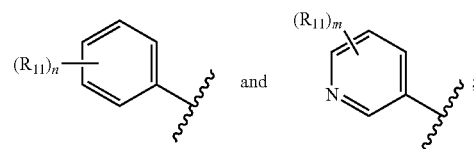
2. The compound of claim 1, wherein the compound of

formula (I) has the following structure:

88



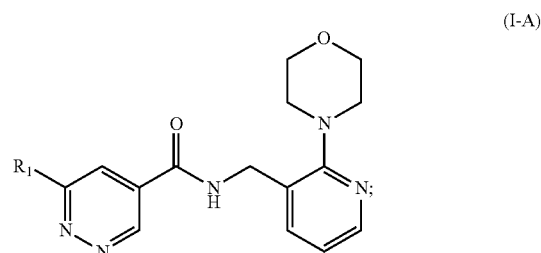
wherein:

 q is an integer selected from 0, 1, 2, 3, 4, 5, 6, 7, and 8; t is an integer selected from 0, 1, 2, and 3;each R_{15} is independently selected from deuterium, hydrogen, C_1 - C_4 alkyl, and $-CF_3$;each R_{16} is independently selected from hydrogen and halogen; R_1 is selected from:

wherein:

 n is an integer selected from 0, 1, 2, 3, 4, and 5; m is an integer selected from 0, 1, 2, 3, and 4; andeach R_{11} is independently selected from the groupconsisting of H, C_1 - C_8 alkyl, halogen, cyano, C_1 - C_8 alkoxy, $-O-(CH_2)_p-R_y$, wherein p is an integerselected from 0, 1, 2, 3, and 4, and R_y is selected from C_3 - C_{10} cycloalkyl and C_1 - C_8 haloalkyl, and $-O-CH(R_{12})-R_{13}$, wherein R_{12} is H or C_1 - C_8 alkyland R_{13} is C_3 - C_8 cycloalkyl.

3. The compound of claim 2, wherein the compound of formula (I) has the following structure:



wherein:

 R_1 is selected from: